

EE 522 - PsiQuantum Group - Weekly Report 9

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Previous Meeting Items

We were implementing joint benchmark plots around existing algorithms based on the several metrics Mariia recommended: number of qubits, Toffoli gates including with “Gidney elbows,” T-gates, and active volume.

Future Plan

We are finalizing the plots for the algorithms we have finished and including pre-existing algorithms (Mariia clarified that they had both in-place and out-of-place implementations so that we can compare them to both sets of adders for a more complete, robust comparison) and finalizing the poster hopefully by tomorrow morning so that Mariia can review it and we can send it for printing.

Problems

Workbench’s AWS hosting still periodically slows down such that terminal commands aren’t going through, it’s not reflecting typing of commands, you can’t look at plots, and because of the prior issues generating plots or pushing them to GitHub isn’t possible, and similarly Jupyter Notebooks can’t find Python environments or execute code cells. Comes up randomly seemingly irrespective of local internet and can lead to frustrating delays.